

MATRIXPOLICY FOR ROLE-AWARE OPTIMIZATION OF RATIONAL TRANSFORMER NONLINEARITIES

Anonymous authors

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ABSTRACT

A useful LLM pretraining recipe moves the loss-time frontier by reaching the same validation loss with fewer tokens or less wall-clock time. We target this objective through the Transformer nonlinear sublayer, treating it as an optimizer-visible interface. MatrixPolicy with Rational Latent Basis (RLB) replaces a fixed scalar response with grouped rational responses that expose local gain, coefficient-activity, and matrix-balance statistics to a role-aware update. MatrixPolicy restricts the signal-conditioned update to the sublayer input and output matrices; rational coefficients and the rest of the Transformer remain on AdamW. Across DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4 with a fixed 12-layer, width-768 decoder, MatrixPolicy reaches matched validation-loss targets with fewer tokens across all ten dataset-budget cells under the same protocol. It saves 19.5–22.0% target-arrival tokens versus SiLU with AdamW and 29.2–34.0% versus SiLU with Muon at 100M tokens, and 22.6–26.4% and 6.4–10.6%, respectively, at 300M tokens. [todo: Insert the larger-scale training result here.]

1 INTRODUCTION

Recent LLM optimizer papers frame practical efficiency through loss-time and compute-time frontiers, asking which training recipe reaches a target loss with fewer tokens, FLOPs, or wall-clock time (Shah et al., 2025; Liu et al., 2025; Semenov et al., 2025). This criterion also exposes an activation-design question. In a modern decoder block, the nonlinear sublayer is a learned matrix–nonlinearity–matrix map. The activation sets local gain and curvature, while the adjacent matrices choose and recombine the coordinates on which that response acts. Standard training pipelines leave this local structure implicit to the optimizer. Activation coefficients and adjacent matrices are updated as ordinary parameters, leaving gain, coefficient-activity, and role information inside the sublayer unused by the matrix update.

We make this interface concrete with Rational Latent Basis (RLB) and MatrixPolicy. RLB maps residual states into grouped latent coordinates, normalizes each group by its RMS radius, and applies a trainable real-pole-free P5/Q4 rational response. The resulting response-gain, coefficient-activity, matrix-pressure, and pair-balance signals are detached from the forward graph and passed to MatrixPolicy. MatrixPolicy restricts their influence to the RLB input and output matrices; attention, embeddings, normalization parameters, rational coefficients, and ordinary Transformer tensors remain on AdamW. This local division exposes the quantities created by the nonlinear sublayer while preserving AdamW for the rest of the model.

We evaluate target arrival by measuring the tokens and wall-clock time required to reach a fixed validation-loss target under the same model, corpus, and protocol. In the completed study, the same 12-layer, width-768 causal Transformer is trained on DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4. The experiments include RLB-only controls and matched SiLU with AdamW and SiLU with Muon baselines under the same model, data, and measurement protocol. MatrixPolicy reaches shared validation-loss targets earlier at both 100M and 300M token budgets. [todo: Insert the larger-scale training result here.] The contribution is an activation design that exposes optimizer-visible matrix roles, a localized role-aware optimizer for the exposed matrix pair, and matched evidence that this interface reduces target-arrival tokens across five pretraining corpora while reporting the corresponding observed wall-clock frontier.

2 RELATED WORK

Prior work on Transformer nonlinear sublayers has developed the projection topology and scalar response along largely separate lines. The topology line begins with multiplicative GLU gates and reaches Transformer sublayers through SwiGLU (Dauphin et al., 2017; Shazeer, 2020). Related gated forms are used from LLaMA through Llama 3 and in recent DeepSeek-V3 and Qwen3 models (Touvron et al., 2023; Grattafiori et al., 2024; DeepSeek-AI et al., 2024; Yang et al., 2025). These architectures learn the surrounding projections while retaining a response family selected before pretraining.

The response line instead makes the scalar map trainable. Rational networks provide efficient smooth-function approximation and recent expressivity separations from fixed and gated Transformer responses (Boullé et al., 2020; Tang & Townsend, 2026). Padé Activation Units established end-to-end learning of rational coefficients, and later work studied learnable rational activations in Transformer pretraining and fine-tuning (Molina et al., 2020; Fang et al., 2023). Within a matrix-response-matrix sublayer, the realized response enters the output-projection gradient and its derivative enters the input-projection gradient, as derived in Appendix C.1. Coefficient learning makes both factors evolve and supplies coefficient-gradient activity as an additional local signal. The rational-activation formulations above retain a base optimizer for the coefficients and adjacent matrices, leaving a response-summary interface outside their scope.

Optimizer work differs primarily in the parameter structure used to transform those gradients. AdamW retains coordinatewise moment adaptation with decoupled weight decay (Loshchilov & Hutter, 2019). Sophia, Adam-mini, and GaLore add diagonal curvature, block-shared rates, or low-rank gradient subspaces (Liu et al., 2024; Zhang et al., 2025; Zhao et al., 2024). SOAP and Muon extend adaptation to matrix geometry through an evolving eigenbasis or orthogonalized momentum (Vyas et al., 2025; Liu et al., 2025). Across these methods, conditioning is derived from gradient history, curvature, and parameter geometry. The nonlinear response that produced the gradients and the input/output roles of its adjacent matrices remain implicit.

MuonClip provides a directly relevant example of module-conditioned optimization. Its maximum attention logit triggers post-update rescaling of the query and key matrices that produced it (Kimi Team et al., 2025). MatrixPolicy applies this module-conditioned viewpoint to the nonlinear sublayer, where detached response statistics condition the two projections surrounding a learned activation. This links learnable activation and matrix-aware optimizer work through a module-specific control interface.

3 METHOD

RLB places trainable grouped rational responses between an input/output matrix pair. Its RMS-normalized form supplies response statistics and a reciprocal pair-scale coordinate. MatrixPolicy uses detached summaries of this local state to redistribute group gradients, shape the two matrix updates, and balance their relative scale. In the evaluated configuration, MatrixPolicy updates only the RLB matrix pair; rational coefficients and all other Transformer parameters use AdamW.

3.1 RATIONAL LATENT BASIS

For layer $l \in \{1, \dots, L\}$, let $x_l \in \mathbb{R}^d$ be the nonlinear sublayer input. RLB uses bias-free matrices $A_l \in \mathbb{R}^{H \times d}$ and $B_l \in \mathbb{R}^{d \times H}$ with $H = Gm$. For $g \in \{1, \dots, G\}$, $A_{l,g} \in \mathbb{R}^{m \times d}$ is a row block of A_l and $B_{l,g} \in \mathbb{R}^{d \times m}$ is the matching column block of B_l . One scalar P5/Q4 curve is shared by all m coordinates in a group,

$$R_{l,g}(u) = \frac{\sum_{i=0}^5 a_{l,g,i} u^i}{1 + \sum_{j=1}^4 |b_{l,g,j}| |u|^j}, \quad u \in \mathbb{R}. \quad (1)$$

The evaluated parameterization is the group-shared curve in equation 1, with ten trainable coefficients per group. The coefficients are initialized from a SiLU fit on $[-5, 5]$ and trained by AdamW. The denominator is at least one on $\mathbb{R}$, so the response has no real poles. Absolute-value derivatives use $\text{sign}(0) = 0$ and $R'_{l,g}$ denotes the scalar derivative almost everywhere.

With the evaluated RMS floor $\epsilon_{\text{rms}} = 10^{-6}$, the complete group map is

$$\begin{aligned}\rho_{l,g}(x_l) &= \sqrt{m^{-1}\|A_{l,g}x_l\|_2^2 + \epsilon_{\text{rms}}}, & u_{l,g}(x_l) &= \frac{A_{l,g}x_l}{\rho_{l,g}(x_l)}, \\ h_{l,g}(x_l) &= \rho_{l,g}(x_l)R_{l,g}(u_{l,g}(x_l)), & y_l &= \sum_{g=1}^G B_{l,g}h_{l,g}(x_l).\end{aligned}\quad (2)$$

The scalar response acts coordinatewise on $u_{l,g}$. The normalization separates curve shape from group radius, while the prefactor returns that radius before B_l recombines the groups.

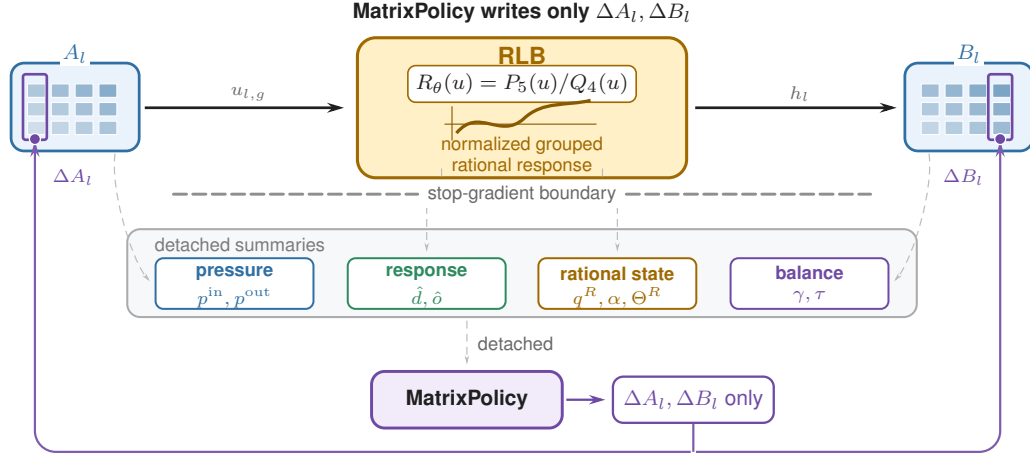


Figure 1: RLB activation-optimizer interface. The matrix A_l forms normalized rational groups and B_l recombines $h_l = \text{concat}_{g=1}^G h_{l,g}$. Detached forward, gradient, coefficient-activity, and pair-balance summaries condition MatrixPolicy updates of (A_l, B_l) ; rational coefficients follow AdamW.

The radius factor also exposes a positive pair coordinate. For fixed $R_l = \{R_{l,g}\}_{g=1}^G$, write equation 2 as $f_l(x; A_l, B_l, R_l)$. If $\epsilon_{\text{rms}} = 0$ and $A_{l,g}x \neq 0$ for every group, then for any $\lambda \in \mathbb{R}_{>0}^G$ and $D_l(\lambda) = \text{diag}(\lambda_1 I_m, \dots, \lambda_G I_m)$,

$$f_l(x; A_l, B_l, R_l) = f_l(x; D_l(\lambda)A_l, B_l D_l(\lambda)^{-1}, R_l). \quad (3)$$

Appendix B derives this positive homogeneity and the floor perturbation. MatrixPolicy uses only bounded moves along the resulting approximate pair coordinate.

3.2 MATRIXPOLICY

MatrixPolicy controls only (A_l, B_l) . Its inputs are detached summaries from the RLB activation, the globally clipped gradients, and fixed-grid curve probes; its outputs are three ordered operations: group-gradient gating, a role/depth matrix step, and reciprocal pair rescaling. No summary tensor enters the forward graph.

Let $t \in \{1, \dots, T\}$ be the one-based optimizer step and $\xi_t = t/T$. Backpropagation is followed by one global gradient clip. The wrapper updates the pressure/activity memories; backbone AdamW then updates the rational coefficients and all non-RLB-matrix tensors. MatrixPolicy scales the same clipped gradients of A_l, B_l , applies AdamW and then Muon, and produces $(A_l^{\text{desc}}, B_l^{\text{desc}})$ without a second global clip. A due probe refresh observes the updated curves, and every fifth step the wrapper measures and rescales the post-descent pair. Figure 2 summarizes this order.

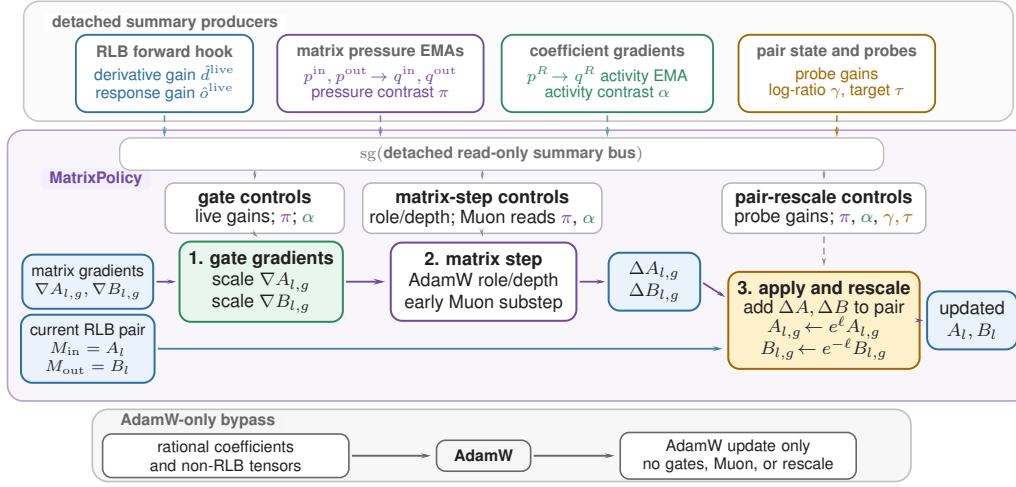


Figure 2: MatrixPolicy signal flow. A read-only stop-gradient bus carries forward, gradient, curve-probe, and post-descent pair summaries to the three controls. Separate lower lanes track the gradient update and current pair; rational coefficients and non-RLB tensors follow AdamW.

Set $M_{l,\text{in}} = A_l$ and $M_{l,\text{out}} = B_l$, with $\sigma \in \{\text{in}, \text{out}\}$. For a tensor V and positive group values v_j , define

$$\begin{aligned} \text{rms}_\epsilon(V) &:= \sqrt{\text{mean}(V^2) + \epsilon}, \\ \text{geomean}_{j=1}^G(v_j) &:= \exp\left(\frac{1}{G} \sum_{j=1}^G \log v_j\right), \\ \text{clip}(x, a, b) &:= \min\{\max\{x, a\}, b\}, \\ \text{smoothstep}(a, b, x) &:= 3\omega^2 - 2\omega^3, \quad \omega = \text{clip}\left(\frac{x-a}{b-a}, 0, 1\right). \end{aligned} \quad (4)$$

We write $\text{rms} = \text{rms}_0$ and use $\epsilon_{\text{live}} = 10^{-6}$ and $\epsilon_p = \epsilon_{\text{probe}} = 10^{-8}$.

All gradient summaries below use the clipped, pre-policy gradients. With $a_{l,g} = (a_{l,g,0}, \dots, a_{l,g,5})$ and $b_{l,g} = (b_{l,g,1}, \dots, b_{l,g,4})$, the instantaneous measurements are

$$p_{l,g}^{\text{in}} = \frac{\text{rms}_{\epsilon_p}(\nabla A_{l,g})}{\text{rms}_{\epsilon_p}(A_{l,g})}, \quad p_{l,g}^{\text{out}} = \frac{\text{rms}_{\epsilon_p}(\nabla B_{l,g})}{\text{rms}_{\epsilon_p}(B_{l,g})}, \quad (5)$$

$$p_{l,g}^R = \left[\frac{1}{2} (\text{mean}((\nabla a_{l,g})^2) + \text{mean}((\nabla b_{l,g})^2)) + \epsilon_p \right]^{1/2}. \quad (6)$$

The first two quantities are relative matrix-gradient proxies. The quantity $p_{l,g}^R$ is the square root of the equally weighted numerator-block and denominator-block gradient mean squares. Current-step detached memories are

$$q_{l,g,t}^r = 0.95 q_{l,g,t-1}^r + 0.05 p_{l,g,t}^r, \quad r \in \{\text{in}, \text{out}, R\}, \quad (7)$$

and are initialized from the first observed values. The controls use

$$\pi_{l,g} = \log q_{l,g}^{\text{in}} - \log q_{l,g}^{\text{out}}, \quad \alpha_{l,g} = \log q_{l,g}^R - \frac{1}{2} (\log q_{l,g}^{\text{in}} + \log q_{l,g}^{\text{out}}). \quad (8)$$

The pressure contrast $\pi_{l,g}$ compares the adjacent matrix roles, whereas $\alpha_{l,g}$ compares the block-weighted coefficient-gradient memory with the geometric mean of the two relative matrix-gradient memories. Appendix C.1 derives the local gradient geometry. We drop t when clear.

The activation also caches per-group live response and derivative gains,

$$\hat{d}_{l,g}^{\text{live}} = \sqrt{\text{mean}(R'_{l,g}(u_{l,g})^2) + \epsilon_{\text{live}}}, \quad \hat{o}_{l,g}^{\text{live}} = \sqrt{\text{mean}(R_{l,g}(u_{l,g})^2) + \epsilon_{\text{live}}}. \quad (9)$$

Each invocation of the RLB module increments the cache counter. On the first invocation and whenever the counter is divisible by eight, the implementation flattens the token positions in their stored order and, when more than 512 are present, retains 512 evenly spaced indices. The means in equation 9 use those positions, all m coordinates, and FP32 arithmetic. Derivative gain supplies the input-role proxy and response gain the output-role proxy; an unavailable cache contributes the identity multiplier.

For pair balancing, let $\mathcal{U} = \{u_k\}_{k=1}^{129}$ be uniform on $[-5, 5]$. The first five-step rescale event with $s_{l,t} > 0$ computes the fixed-grid gains; a due event refreshes them once at least ten optimizer steps have elapsed,

$$\hat{d}_{l,g}^{\text{probe}} = \sqrt{\frac{1}{129} \sum_{u_k \in \mathcal{U}} R'_{l,g}(u_k)^2 + \epsilon_{\text{probe}}}, \quad \hat{o}_{l,g}^{\text{probe}} = \sqrt{\frac{1}{129} \sum_{u_k \in \mathcal{U}} R_{l,g}(u_k)^2 + \epsilon_{\text{probe}}}. \quad (10)$$

Immediately before a due reciprocal rescale, the post-descent pair supplies

$$\gamma_{l,g} = \log \text{rms}_{\epsilon_p}(A_{l,g}^{\text{desc}}) - \log \text{rms}_{\epsilon_p}(B_{l,g}^{\text{desc}}). \quad (11)$$

The probes summarize curve shape independently of the sampled live gains; the third control combines them and $\gamma_{l,g}$ with $\pi_{l,g}, \alpha_{l,g}$.

3.2.1 GROUP-GRADIENT GATING

The first control redistributes gradient scale among groups inside each matrix role. It reads the sampled live gains, the pressure contrast $\pi_{l,g}$, and the activity contrast $\alpha_{l,g}$, then produces scaled gradients for A_l and B_l . A smooth gate $\phi_t = \text{smoothstep}(0.02, 0.30, \xi_t)$ turns on this control in the reported runs. The evaluated manifest sets gain, pressure, and activity strengths to 0.20, 0.10, and 0.20, and clips group multipliers to $[0.75, 1.35]$. The group-gradient rule is

$$\begin{aligned} \tilde{\pi}_{l,g} &= \text{clip}(\pi_{l,g}, -1.50, 1.50), & c_{l,g}^{\text{activity}} &= \exp\left[-0.20\phi_t \text{ReLU}\left(\frac{\alpha_{l,g} - 0.05}{0.45}\right)\right], \\ c_{l,g,\text{in}}^{\text{gain}} &= \left(\frac{\text{geomean}_{j=1}^G \hat{d}_{l,j}^{\text{live}}}{\hat{d}_{l,g}^{\text{live}}}\right)^{0.20\phi_t}, & c_{l,g,\text{out}}^{\text{gain}} &= \left(\frac{\text{geomean}_{j=1}^G \hat{o}_{l,j}^{\text{live}}}{\hat{o}_{l,g}^{\text{live}}}\right)^{0.20\phi_t}, \\ c_{l,g,\text{in}}^{\text{pressure}} &= \exp(-0.10\phi_t \tilde{\pi}_{l,g}), & c_{l,g,\text{out}}^{\text{pressure}} &= \exp(+0.10\phi_t \tilde{\pi}_{l,g}), \\ c_{l,g,\sigma}^{\text{raw}} &= c_{l,g,\sigma}^{\text{gain}} c_{l,g,\sigma}^{\text{pressure}} c_{l,g}^{\text{activity}}, & c_{l,g,\sigma} &= \text{clip}\left(\frac{c_{l,g,\sigma}^{\text{raw}}}{\text{geomean}_{j=1}^G c_{l,j,\sigma}^{\text{raw}}}, 0.75, 1.35\right). \end{aligned} \quad (12)$$

Live derivative gain controls the input role, live response gain controls the output role, and pressure enters the two raw role multipliers with opposite signs. Before recentering, positive $\pi_{l,g}$ contributes a smaller input multiplier and a larger output multiplier, while high coefficient activity contributes shared damping. Geometric recentering makes the final effect relative to the layerwise mean combined control; clipping can shift that mean. Appendix C.2 gives the exact log-space identity.

The scaled matrix gradients are

$$\tilde{\nabla} A_{l,g} = c_{l,g,\text{in}} \nabla A_{l,g}, \quad \tilde{\nabla} B_{l,g} = c_{l,g,\text{out}} \nabla B_{l,g}. \quad (13)$$

Let $\tilde{\nabla} M_{l,\text{in}}$ be the row-block concatenation of $\{\tilde{\nabla} A_{l,g}\}_{g=1}^G$, and let $\tilde{\nabla} M_{l,\text{out}}$ be the column-block concatenation of $\{\tilde{\nabla} B_{l,g}\}_{g=1}^G$. These scaled gradients pass directly to the matrix optimizers without another global-norm clip.

3.2.2 ROLE/DEPTH MATRIX STEP

The second control applies the matrix optimizer step to the scaled gradients. It assigns different depth trends to the two roles, applies AdamW to both matrices, and adds a transient Muon substep. Let $\delta_l = (l-1)/(L-1)$ for $L > 1$ and $\delta_l = 1/2$ otherwise. The role/depth factors are

$$\begin{aligned} \varrho_{\text{in}}(l) &= \text{clip}(1 - 0.50(\delta_l - 0.5), 0.55, 1.40), \\ \varrho_{\text{out}}(l) &= \text{clip}(1 + 1.00(\delta_l - 0.5), 0.55, 1.40), \\ s_{l,\sigma}^{\text{Adam}} &= \text{clip}(3.0 \max\{0.10, 1 + 1.20(\varrho_{\sigma}(l) - 1)\}, 0.40, 4.0). \end{aligned} \quad (14)$$

The pressure and activity memories attenuate only the Muon learning rate in this control. Their layer summaries and transient schedule are

$$\begin{aligned}
\Psi_{l,t} &= \text{clip}\left(G^{-1} \sum_{g=1}^G |\pi_{l,g}|, 0, 1.50\right), \\
\Omega_{l,t} &= \text{ReLU}\left(\frac{G^{-1} \sum_{g=1}^G \alpha_{l,g} - 0.05}{0.45}\right), \\
\psi_{l,t} &= \text{clip}(\exp[-0.30\Psi_{l,t} - 0.65\Omega_{l,t}], 0.10, 1.15), \\
\omega_t^\mu &= \text{smoothstep}(0.02, 0.12, \xi_t)[1 - \text{smoothstep}(0.20, 0.36, \xi_t)], \\
\mu_{l,\sigma,t} &= \text{clip}(0.75 \omega_t^\mu \varrho_\sigma(l) \psi_{l,t}, 0, 0.75).
\end{aligned} \tag{15}$$

Thus $\mu_{l,\sigma,t}$ is largest inside the early window when role imbalance and coefficient activity are small. With base learning rate η_t and evaluated Muon scale $s_\mu = 1$, define $\eta_{l,\sigma,t}^A = \eta_t s_{l,\sigma}^{\text{Adam}}(1 - \mu_{l,\sigma,t})$ and $\eta_{l,\sigma,t}^\mu = \eta_t s_\mu \mu_{l,\sigma,t}$. AdamW and Muon consume the same scaled gradient, retain separate states, and act sequentially:

$$M_{l,\sigma}^A = \text{AdamW}_{\eta_{l,\sigma,t}^A}(M_{l,\sigma}^t; \tilde{M}_{l,\sigma}^t), \quad M_{l,\sigma}^{\text{desc}} = \text{Muon}_{\eta_{l,\sigma,t}^\mu}(M_{l,\sigma}^A; \tilde{M}_{l,\sigma}^t). \tag{16}$$

The matrix AdamW uses $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$; both matrix optimizers receive weight decay 0.1. Muon uses the PyTorch Nesterov form with momentum 0.95, five Newton–Schulz steps, and match-RMS learning-rate adjustment (Liu et al., 2025). Before $\xi_t = 0.02$, its zero learning rate prewarms the Muon momentum without moving the parameters; after $\xi_t \geq 0.36$, the Muon optimizer is skipped. Appendix C.3 gives the corresponding matrix-spectrum geometry, and Appendix A.1 specifies η_t .

3.2.3 RECIPROCAL PAIR RESCALING

The third control acts every five optimizer steps on the post-descent pair. It combines the fixed-grid gains, $\gamma_{l,g}$, and the pressure/activity memories. The target joins the probe-gain balance coordinate derived in Appendix C.4 with a bounded pressure offset:

$$\begin{aligned}
\bar{\pi}_{l,g} &= \text{clip}(\pi_{l,g}, -1.25, 1.25), \\
\tau_{l,g} &= \frac{1}{2} \left(\log \hat{d}_{l,g}^{\text{probe}} - \log \hat{o}_{l,g}^{\text{probe}} \right) + 0.25 \bar{\pi}_{l,g}, \\
\bar{\alpha}_{l,g} &= \text{clip}(\alpha_{l,g}, -1.25, 1.25), \\
\chi_{l,g}^{\text{rs}} &= \exp(0.10 \bar{\alpha}_{l,g}), \\
s_{l,t} &= 0.50 \text{smoothstep}(0.03, 0.35, \xi_t) \text{clip}(1 + 0.15(\delta_l - 0.5), 0.75, 1.25).
\end{aligned} \tag{17}$$

Here $\chi_{l,g}^{\text{rs}}$ controls rescale speed. The clipped log step and final pair are

$$\begin{aligned}
\ell_{l,g,t} &= \text{clip}\left(\frac{1}{2}[\tau_{l,g} - \gamma_{l,g}]s_{l,t}\chi_{l,g}^{\text{rs}}, -0.030, 0.030\right), \\
A_{l,g}^{t+1} &= e^{\ell_{l,g,t}} A_{l,g}^{\text{desc}}, \quad B_{l,g}^{t+1} = e^{-\ell_{l,g,t}} B_{l,g}^{\text{desc}}.
\end{aligned} \tag{18}$$

The first term of $\tau_{l,g}$ balances the surrogate role pressures $\hat{d}_{l,g}^{\text{probe}} e^{-\gamma_{l,g}}$ and $\hat{o}_{l,g}^{\text{probe}} e^{\gamma_{l,g}}$; reciprocal matrix scaling leaves both probe gains fixed. The pressure term shifts the target, while $\chi_{l,g}^{\text{rs}}$ sets the clipped rescale speed. Thus larger $\alpha_{l,g}$ accelerates pair balancing even as the earlier controls reduce gradient-driven matrix motion. The move preserves the nonzero, unfloored map in equation 3. The evaluated activation floor makes this approximate, and the scale clip gives no relative-error guarantee at the floor. On other steps, $(A_l^{t+1}, B_l^{t+1}) = (A_l^{\text{desc}}, B_l^{\text{desc}})$. Only the matrix parameters are rescaled; nested AdamW and Muon state buffers remain in their pre-rescale coordinates.

4 EXPERIMENTS

4.1 FIXED-SCALE TARGET-ARRIVAL STUDY

Recent optimizer studies show that practical rankings depend on compute-time accounting, controlled tuning, standardized pretraining scenarios, and model or data scale (Shah et al., 2025; Semenov et al., 2025; Wen et al., 2026; Volkova et al., 2026). The fixed-scale study therefore uses matched models, data, budgets, and loss targets while reporting both token and wall-clock arrival.

The fixed-scale study uses the same 12-layer, width-768 causal Transformer on five corpora: DCLM/DataComp-LM (Li et al., 2024), FineWeb-Edu and FineWeb (Penedo et al., 2024), Dolma-sample (Soldaini et al., 2024), and C4 (Raffel et al., 2020). The 100M-token budget uses 3050 optimizer steps and the 300M-token budget uses 9150 optimizer steps, with three seeds per method/dataset pair. The main comparison keeps the architecture and data fixed while changing the activation and optimizer used in the nonlinear sublayer: MatrixPolicy is compared with SiLU controls and with RLB-only controls, where RLB-only places the same activation under the named standard optimizer. Shared optimizer, evaluation-cadence, and throughput-measurement details are reported in Appendix A.1; broader optimizer comparisons beyond AdamW and Muon are summarized in Table 2.

Table 1: Target-arrival efficiency for the fixed 12-layer, width-768 Transformer. RLB denotes the same rational latent-basis activation without MatrixPolicy. At the hardest validation-loss target reached by every listed method in all three seeds, each token-saving column reports MatrixPolicy’s savings relative to the named method, followed by observed time to target for every method.

Budget	Dataset	Target loss	MatrixPolicy token savings (%)				Time to target (min)				
			vs. SiLU AdamW	vs. SiLU Muon	vs. RLB AdamW	vs. RLB Muon	Matrix Policy	SiLU AdamW	SiLU Muon	RLB AdamW	RLB Muon
100M	DCLM	4.55	20.7	32.4	20.7	34.3	12.7	20.8	26.0	14.3	19.0
	FineWeb-Edu	4.40	19.5	29.5	20.2	29.5	11.7	20.3	24.4	14.2	18.0
	FineWeb	4.60	22.0	31.4	20.7	31.4	13.3	16.4	20.2	16.2	20.1
	Dolma	4.60	22.0	34.0	22.7	34.9	13.4	17.6	22.0	17.3	22.2
	C4	4.60	20.7	29.2	19.3	28.7	12.8	14.3	17.3	14.9	18.8
300M	DCLM	4.10	23.3	8.3	24.0	7.0	42.1	53.5	45.9	54.7	45.6
	FineWeb-Edu	3.85	22.6	6.4	22.4	4.5	48.3	59.9	47.9	52.3	50.0
	FineWeb	4.10	22.9	6.8	22.1	5.0	46.2	61.2	46.4	55.6	46.6
	Dolma	3.95	23.0	10.6	23.0	7.3	45.6	49.3	52.3	50.5	47.4
	C4	4.00	26.4	8.5	25.3	6.5	48.3	64.7	55.5	59.4	51.9

Table 1 is the primary empirical result. It asks: for the hardest common validation-loss target reached by every listed method in all three seeds, what fraction of target-arrival tokens does MatrixPolicy save, and how many measured target-arrival minutes does each method require? Across all five datasets under the 100M-token budget, MatrixPolicy saves 19.5–22.0% of the SiLU with AdamW target-arrival tokens and 29.2–34.0% of the SiLU with Muon target-arrival tokens. Under the 300M-token budget, after longer training compresses endpoint gaps, MatrixPolicy still saves 22.6–26.4% versus SiLU with AdamW and 6.4–10.6% versus SiLU with Muon. Its observed target-arrival time is lower in 39 of the 40 listed comparisons. Because node assignments were not matched, the time columns describe the realized allocations rather than a hardware-normalized optimizer comparison.

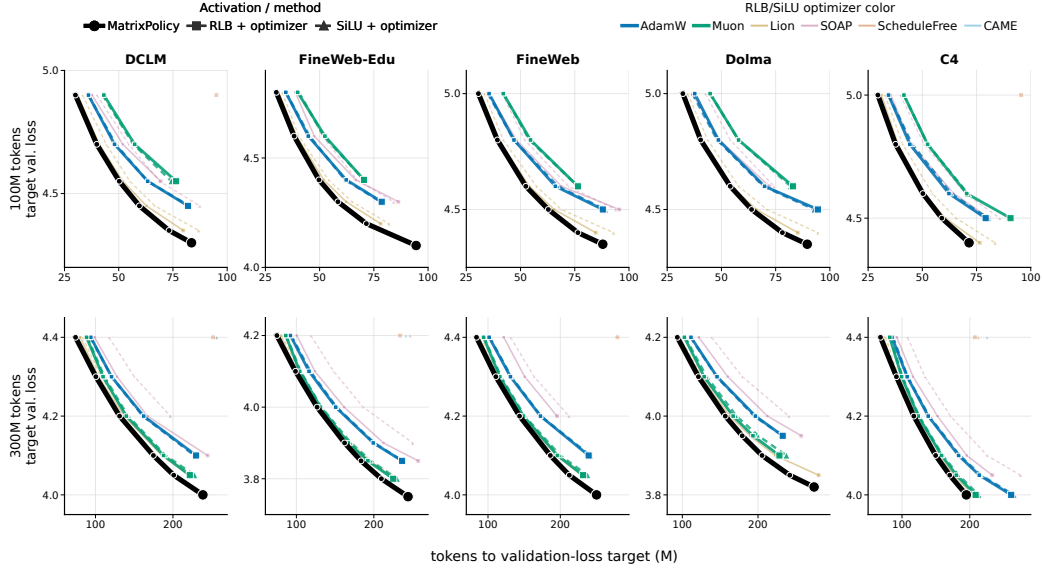


Figure 3: MatrixPolicy shifts target arrival left across the completed fixed-scale study. Rows show the 100M- and 300M-token budgets, and columns show datasets. Each point is the mean token count needed to reach one validation-loss target in all three seeds; connected points trace one method from easier to harder targets. Black circles are MatrixPolicy. For every optimizer color, solid squares are RLB with that optimizer and dashed triangles are SiLU with the same optimizer. Lower targets are harder, and the hard-target timing and token-saving percentages are quantified in Table 1.

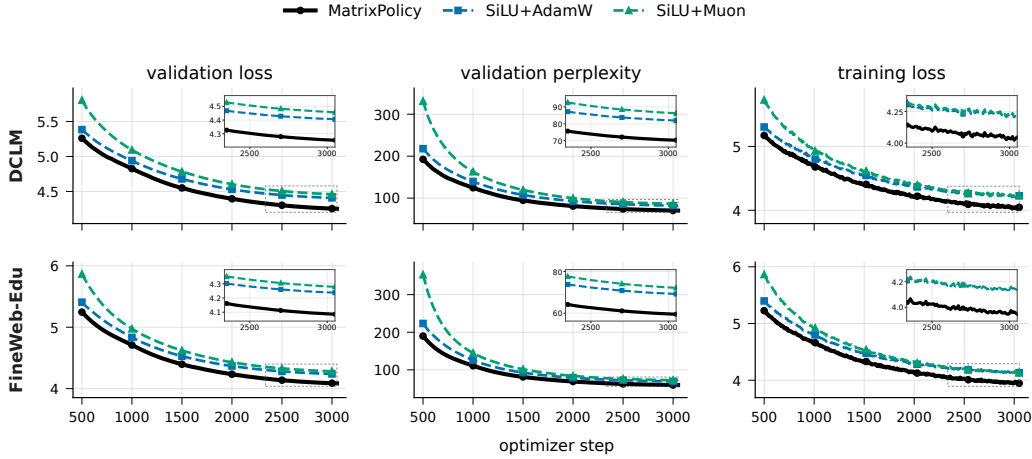


Figure 4: Representative 100M-token-budget training dynamics on DCLM and FineWeb-Edu. Rows show datasets and columns show validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. This figure uses the two standard SiLU baselines so that the early target-arrival effect in Table 1 remains visually legible; inset windows zoom the final-quarter saturation region.

The 100M-token-budget curves show why target arrival is the right main readout. MatrixPolicy separates early in validation loss and validation perplexity while keeping training loss competitive. A long fixed budget eventually compresses many activation/optimizer endpoints, especially under the 300M-token budget, so the paper emphasizes the number of tokens and minutes needed to reach the same useful validation-loss levels.

4.2 TODO: LARGER-SCALE TRAINING RESULTS

TODO: Insert the larger-scale training result here.

4.3 TODO: COMPONENT ABLATIONS

TODO: Insert the component ablation study here.

5 CONCLUSION

RLB turns the nonlinear Transformer sublayer into an optimizer-visible interface by exposing grouped normalized coordinates, trainable groupwise rational responses, and an idealized positive matrix-pair relation used as a bounded balancing coordinate. MatrixPolicy uses that interface to update the RLB input and output matrices with role-aware statistics, while rational coefficients and non-RLB-matrix parameters use AdamW. The completed 100M-token and 300M-token runs show that MatrixPolicy reaches common validation losses with fewer tokens across every listed dataset-budget cell and lower observed target-arrival time in 39 of 40 listed comparisons. [todo: Complete this conclusion after the larger-scale and ablation results are added.]

REFERENCES

- Nicolas Boullé, Yuji Nakatsukasa, and Alex Townsend. Rational Neural Networks. In *Advances in Neural Information Processing Systems*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/a3f390d88e4c41f2747bfa2f1b5f87db-Abstract.html>.
- Yann N. Dauphin, Angela Fan, Michael Auli, and David Grangier. Language Modeling with Gated Convolutional Networks. In *Proceedings of the 34th International Conference on Machine Learning*, pp. 933–941, 2017. URL <https://proceedings.mlr.press/v70/dauphin17a.html>.
- DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chengqi Deng, Chong Ruan, Damai Dai, et al. DeepSeek-V3 Technical Report. *arXiv preprint arXiv:2412.19437*, 2024. URL <https://arxiv.org/abs/2412.19437>.
- Haishuo Fang, Ji-Ung Lee, Nafise Sadat Moosavi, and Iryna Gurevych. Transformers with Learnable Activation Functions. In *Findings of the Association for Computational Linguistics: EACL 2023*, 2023. URL <https://aclanthology.org/2023.findings-eacl.181/>.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Kimi Team, Yifan Bai, Yiping Bao, Y. Charles, Cheng Chen, Guanduo Chen, Haiting Chen, Huarong Chen, Jiahao Chen, Ningxin Chen, et al. Kimi K2: Open Agentic Intelligence. *arXiv preprint arXiv:2507.20534*, 2025. URL <https://arxiv.org/abs/2507.20534>.
- Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*, 2015. URL <https://mlanthology.org/iclr/2015/kingma2015iclr-adam/>.
- Jeffrey Li, Alex Fang, Georgios Smyrnis, Maor Ivgi, Matt Jordan, Samir Gadre, Hritik Bansal, Etash Guha, et al. DataComp-LM: In Search of the Next Generation of Training Sets for Language Models. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024. doi: 10.52202/079017-0455. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/19e4ea30dded58259665db375885e412-Abstract-Datasets_and_Benchmarks_Track.html.

- Hong Liu, Zhiyuan Li, David Hall, Percy Liang, and Tengyu Ma. Sophia: A Scalable Stochastic Second-order Optimizer for Language Model Pre-training. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/06960915ba8674c7a898ec0b472b80ff-Abstract-Conference.html.
- Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, Yanru Chen, Huabin Zheng, Yibo Liu, Shaowei Liu, Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon is Scalable for LLM Training. *arXiv preprint arXiv:2502.16982*, 2025. URL <https://arxiv.org/abs/2502.16982>.
- Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations*, 2019. URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- Alejandro Molina, Patrick Schramowski, and Kristian Kersting. Padé Activation Units: End-to-End Learning of Flexible Activation Functions in Deep Networks. In *International Conference on Learning Representations*, 2020. URL <https://openreview.net/forum?id=BJlBSkHtDS>.
- Guilherme Penedo, Hynek Kydlicek, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024. doi: 10.52202/079017-0970. URL https://papers.nips.cc/paper/2024/hash/370df50ccfd8bde18f8f9c2d9151bda-Abstract-Datasets_and_Benchmarks_Track.html.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL <http://jmlr.org/papers/v21/20-074.html>.
- Andrei Semenov, Matteo Pagliardini, and Martin Jaggi. Benchmarking Optimizers for Large Language Model Pretraining. *arXiv preprint arXiv:2509.01440*, 2025. URL <https://arxiv.org/abs/2509.01440>.
- Ishaan Shah, Anthony M. Polloreno, Karl Stratos, Philip Monk, Adarsh Chaluvvaraju, Andrew Hojel, Andrew Ma, Anil Thomas, Ashish Tanwer, Darsh J. Shah, Khoi Nguyen, Kurt Smith, Michael Callahan, Michael Pust, Mohit Parmar, Peter Rushton, Platon Mazarakis, Ritvik Kapila, Saurabh Srivastava, Somanshu Singla, Tim Romanski, Yash Vanjani, and Ashish Vaswani. Practical Efficiency of Muon for Pretraining. *arXiv preprint arXiv:2505.02222*, 2025. URL <https://arxiv.org/abs/2505.02222>.
- Noam Shazeer. GLU Variants Improve Transformer. *arXiv preprint arXiv:2002.05202*, 2020. URL <https://arxiv.org/abs/2002.05202>.
- Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin Schwenk, David Atkinson, Russell Authur, Ben Bogin, Khyathi Chandu, et al. Dolma: An Open Corpus of Three Trillion Tokens for Language Model Pretraining Research. *arXiv preprint arXiv:2402.00159*, 2024. URL <https://arxiv.org/abs/2402.00159>.
- Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced Transformer with Rotary Position Embedding. *arXiv preprint arXiv:2104.09864*, 2021. URL <https://arxiv.org/abs/2104.09864>.
- Maosen Tang and Alex Townsend. Rational Neural Networks have Expressivity Advantages. *arXiv preprint arXiv:2602.12390*, 2026. doi: 10.48550/arXiv.2602.12390. URL <https://arxiv.org/abs/2602.12390>.

- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, et al. LLaMA: Open and Efficient Foundation Language Models. *arXiv preprint arXiv:2302.13971*, 2023. URL <https://arxiv.org/abs/2302.13971>.
- Alexandra Volkova, Mher Safaryan, Christoph H. Lampert, and Dan Alistarh. Towards Robust Scaling Laws for Optimizers. *arXiv preprint arXiv:2602.07712*, 2026. URL <https://arxiv.org/abs/2602.07712>.
- Nikhil Vyas, Depen Morwani, Rosie Zhao, Itai Shapira, David Brandfonbrener, Lucas Janson, and Sham Kakade. SOAP: Improving and Stabilizing Shampoo using Adam for Language Modeling. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/e988664070e9591f93fdcf605f7dc623-Abstract-Conference.html.
- Kaiyue Wen, David Leo Wright Hall, Tengyu Ma, and Percy Liang. Fantastic Pretraining Optimizers and Where to Find Them. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=2J51qUZ0iG>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Biao Zhang and Rico Sennrich. Root Mean Square Layer Normalization. *arXiv preprint arXiv:1910.07467*, 2019. URL <https://arxiv.org/abs/1910.07467>.
- Yushun Zhang, Congliang Chen, Ziniu Li, Tian Ding, Chenwei Wu, Diederik P. Kingma, Yinyu Ye, Zhi-Quan Luo, and Ruoyu Sun. Adam-mini: Use Fewer Learning Rates To Gain More. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/45ae878717399e6f62d57c65f052cd46-Abstract-Conference.html.
- Jiawei Zhao, Zhenyu Zhang, Beidi Chen, Zhangyang Wang, Anima Anandkumar, and Yuandong Tian. GaLore: Memory-Efficient LLM Training by Gradient Low-Rank Projection. In *International Conference on Machine Learning*, 2024. URL <https://openreview.net/forum?id=hYHsrKDix7>.

A EXPERIMENT APPENDIX

This appendix gives the fixed-scale protocol behind Section 4.1, the full training curves for the five corpora, and the broader optimizer endpoint comparison used to contextualize the main SiLU and RLB controls.

A.1 FIXED-SCALE TARGET-ARRIVAL STUDY

A.1.1 MODEL AND DATA PIPELINE

The fixed-scale experiments use a custom causal decoder Transformer. The block uses a LLaMA-style pre-normalized layout with RMSNorm before attention and before the nonlinear sublayer, bias-free query/key/value and output attention projections, rotary position embeddings, causal scaled-dot-product attention, residual attention and nonlinear-sublayer updates, a final RMSNorm, and a tied token-embedding/output head (Zhang & Sennrich, 2019; Su et al., 2021; Touvron et al., 2023). The fixed model has 12 layers, width 768, 12 heads of dimension 64, and context length 256.

The SiLU controls use a SwiGLU-like gated nonlinear sublayer with bias-free matrices $W_g, W_v \in \mathbb{R}^{768 \times 2048}$ and $W_o \in \mathbb{R}^{2048 \times 768}$, computing $(\text{SiLU}(xW_g) \odot xW_v)W_o$. MatrixPolicy and RLB-only configurations use a matched single-branch matrix pair. An input matrix maps width 768 to 3072 latent coordinates, the coordinates are partitioned into groups and transformed by the real-pole-free P5/Q4 rational response, and an output matrix maps the 3072 coordinates back to width 768. With the default group size 256, this gives $G = 12$ groups of width $m = 256$; the RMS floor

is $\epsilon_{\text{rms}} = 10^{-6}$. This is the RLB configuration used by the main MatrixPolicy and RLB-control results.

Every rational group starts from the same fixed SiLU-fit coefficients

$$\begin{aligned} a^{(0)} &= (2.4915715 \times 10^{-5}, 0.50006687, 0.24994419, \\ &\quad 0.052620091, 0.0055257656, 0.00024199321), \\ b^{(0)} &= (0.00022554018, 0.10511420, 2.8656992 \times 10^{-5}, 0.00048169621). \end{aligned} \quad (19)$$

Tokenization uses the GPT-2 tokenizer through the Hugging Face tokenizer interface with its base vocabulary; an EOS or pad token is appended between documents. Training samples random contiguous blocks of length 257 and forms next-token prediction pairs of length 256. The DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4 runs use the text field specified in the manifests, with validation slices drawn from held-out token offsets of the same corpus stream.

A.1.2 TRAINING PROTOCOL

All runs use batch size 16, gradient accumulation 2, and therefore 32768 global tokens per optimizer step. The 100M-token budget consists of 3050 optimizer steps; the 300M-token budget consists of 9150 optimizer steps. Each method/dataset pair is run with seeds 1337, 2027, and 3407. Validation is performed every 50 optimizer steps, so token-to-target arrivals are quantized in increments of 1.64M tokens.

The MatrixPolicy rows in the main table use the group-shared P5/Q4 RLB response in equation 1. The evaluated configuration uses AdamW as the backbone optimizer and disables the separate rational-coefficient optimizer. It enables MatrixPolicy group redistribution with gain/pressure/activity strengths 0.20/0.10/0.20, schedule window 0.02–0.30, and group multiplier clip $[0.75, 1.35]$. The RLB matrix pair then uses the role/depth-scaled AdamW branch and the early Muon substep from Section 3.2. The bounded RLB pair-rescale wrapper runs every five optimizer steps after the matrix branch. Rational coefficients, attention weights, embeddings, normalization parameters, and ordinary Transformer weights use AdamW in this configuration.

The base learning rate warms linearly from zero to 3×10^{-4} over 200 optimizer steps and then follows cosine decay to 3×10^{-5} . MatrixPolicy and backbone AdamW use $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The reported DDP runs apply decoupled weight decay 0.1 to two-dimensional backbone parameters, including the tied token embedding/output matrix, and zero decay to rational coefficients and normalization parameters.

Target-arrival minutes in Table 1 are computed seed-wise from the first validation event that reaches the shared target and the same run’s measured seconds per optimizer step. The per-step time is the training-script wall time recorded in the run summary divided by completed optimizer steps for that run; target arrivals are therefore measured with the same evaluation cadence and timing source across methods.

A.1.3 SUPPORTING FIXED-SCALE CURVES

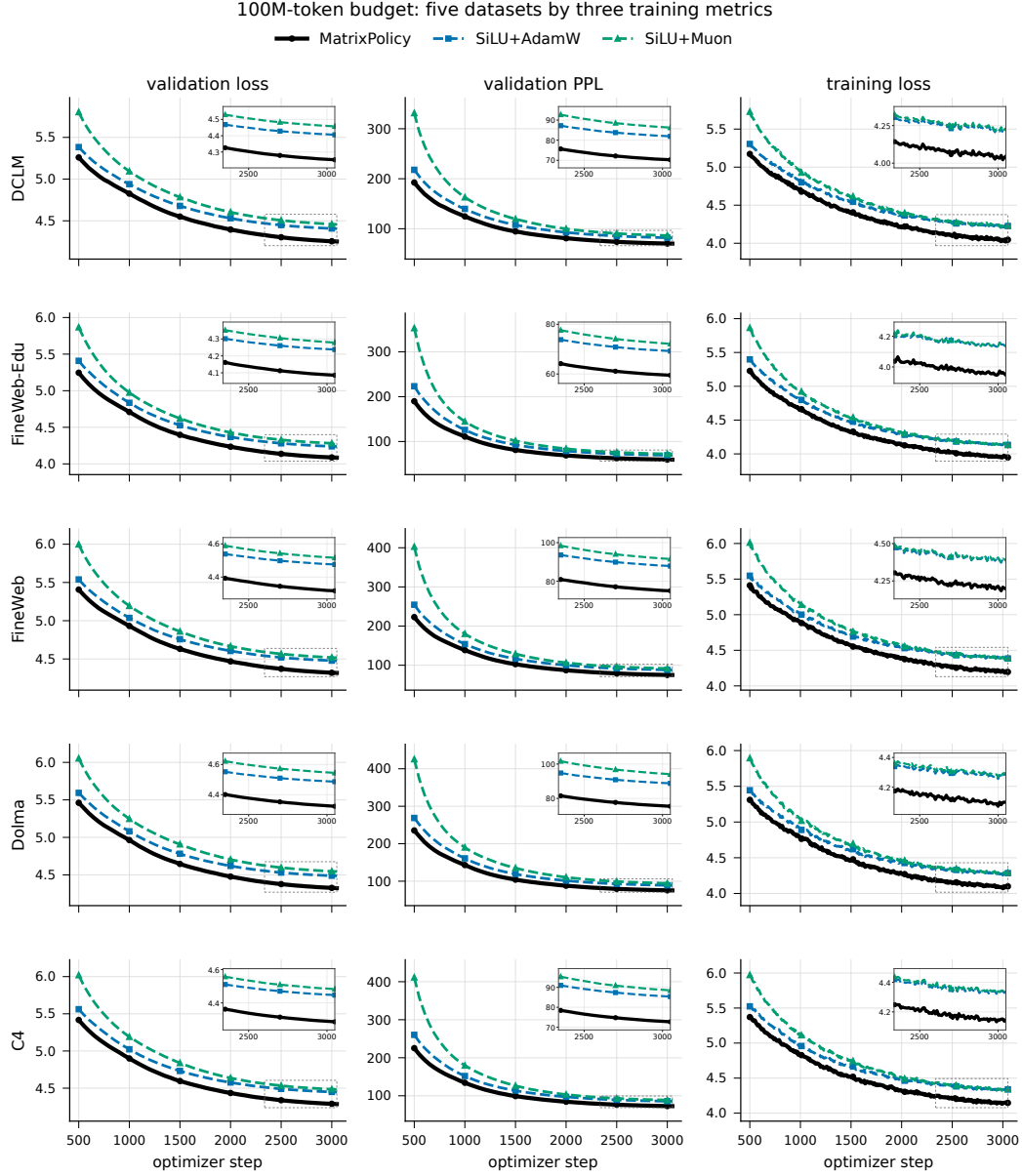


Figure 5: 100M-token-budget dynamics across all five datasets. Rows are datasets and columns are validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. Insets zoom the final-quarter saturation region in each panel.

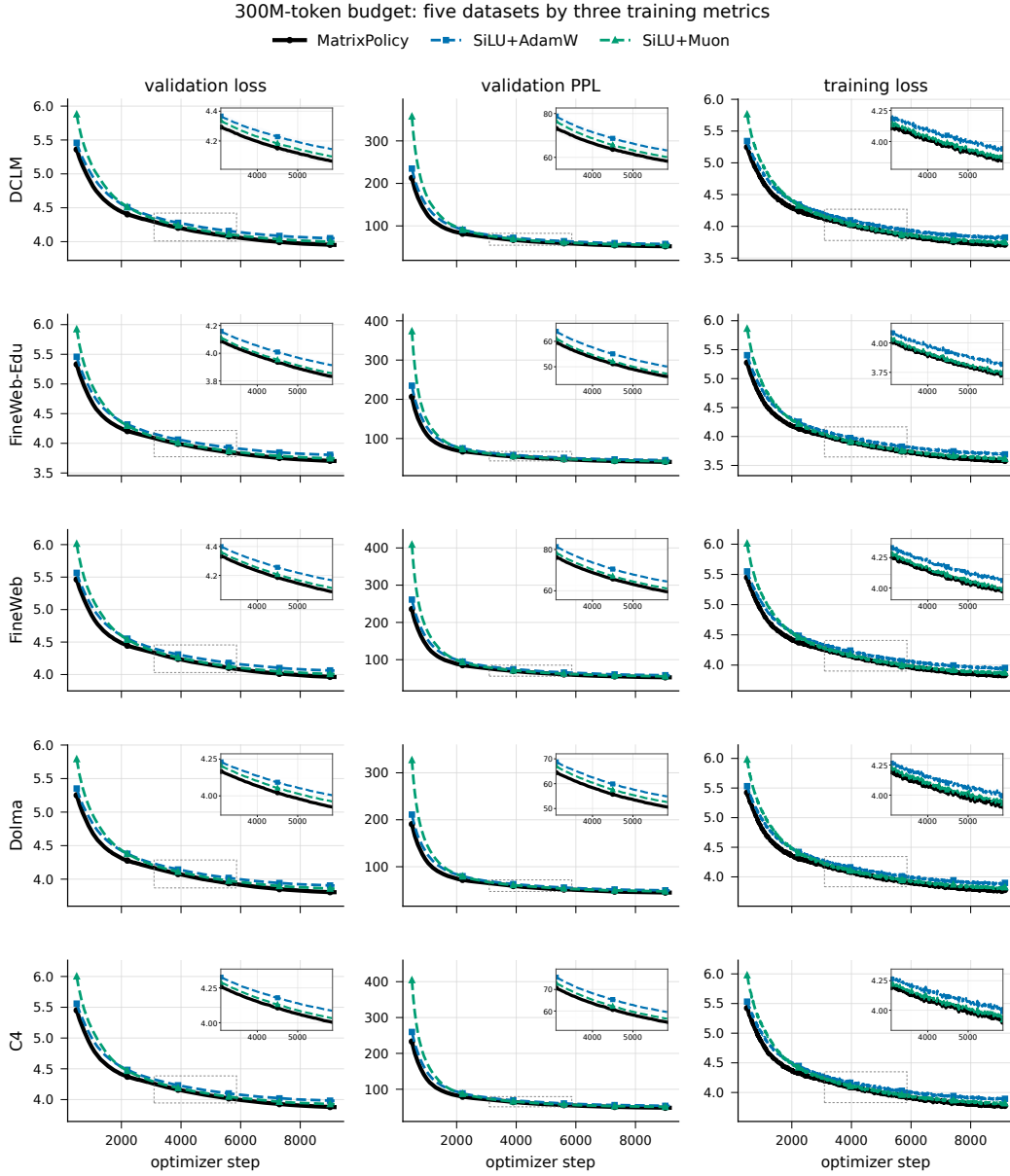


Figure 6: 300M-token-budget dynamics across all five datasets. Rows are datasets and columns are validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. Insets zoom the middle training window where early saturation differences are most visible.

A.1.4 BROAD OPTIMIZER ENDPOINT TABLE

The fixed-scale package also includes the broader optimizer endpoints referenced from Section 4.1. These rows use the same model, datasets, seeds, token budgets, and validation schedule as the main comparison, while replacing the nonlinear-sublayer optimizer according to the table entry.

Table 2: Final validation loss for the broader optimizer sweep on the fixed 12-layer, width-768 language model. Values are mean $\pm$ sample standard deviation over three seeds; lower is better. RLB rows use the same rational latent-basis activation without MatrixPolicy.

Budget	Method	DCLM	FineWeb-Edu	FineWeb	Dolma	C4
100M tokens	MatrixPolicy	4.252 $\pm$ 0.006	4.086 $\pm$ 0.010	4.317 $\pm$ 0.013	4.323 $\pm$ 0.004	4.286 $\pm$ 0.020
	SiLU + AdamW	4.406 $\pm$ 0.010	4.237 $\pm$ 0.009	4.476 $\pm$ 0.010	4.486 $\pm$ 0.001	4.447 $\pm$ 0.016
	SiLU + Lion	4.318 $\pm$ 0.007	4.149 $\pm$ 0.009	4.383 $\pm$ 0.008	4.388 $\pm$ 0.005	4.354 $\pm$ 0.016
	SiLU + Muon	4.457 $\pm$ 0.013	4.279 $\pm$ 0.024	4.516 $\pm$ 0.026	4.544 $\pm$ 0.011	4.482 $\pm$ 0.023
	SiLU + SOAP	4.416 $\pm$ 0.004	4.263 $\pm$ 0.022	4.484 $\pm$ 0.011	4.499 $\pm$ 0.004	4.458 $\pm$ 0.015
	SiLU + ScheduleFree	4.902 $\pm$ 0.011	4.826 $\pm$ 0.007	5.014 $\pm$ 0.019	5.049 $\pm$ 0.015	5.006 $\pm$ 0.016
	SiLU + CAME	5.011 $\pm$ 0.014	4.921 $\pm$ 0.012	5.133 $\pm$ 0.013	5.165 $\pm$ 0.019	5.123 $\pm$ 0.018
	RLB + AdamW	4.405 $\pm$ 0.008	4.239 $\pm$ 0.008	4.472 $\pm$ 0.014	4.488 $\pm$ 0.003	4.441 $\pm$ 0.016
	RLB + Lion	4.295 $\pm$ 0.008	4.136 $\pm$ 0.008	4.363 $\pm$ 0.011	4.362 $\pm$ 0.007	4.327 $\pm$ 0.016
	RLB + Muon	4.467 $\pm$ 0.012	4.283 $\pm$ 0.028	4.516 $\pm$ 0.006	4.548 $\pm$ 0.015	4.476 $\pm$ 0.010
	RLB + SOAP	4.436 $\pm$ 0.044	4.275 $\pm$ 0.024	4.689 $\pm$ 0.336	4.518 $\pm$ 0.036	4.446 $\pm$ 0.019
	RLB + ScheduleFree	4.887 $\pm$ 0.005	4.796 $\pm$ 0.013	4.999 $\pm$ 0.019	5.036 $\pm$ 0.013	4.984 $\pm$ 0.016
	RLB + CAME	5.014 $\pm$ 0.009	4.915 $\pm$ 0.006	5.134 $\pm$ 0.016	5.151 $\pm$ 0.017	5.111 $\pm$ 0.016
300M tokens	MatrixPolicy	3.952 $\pm$ 0.029	3.702 $\pm$ 0.020	3.963 $\pm$ 0.008	3.805 $\pm$ 0.008	3.878 $\pm$ 0.015
	SiLU + AdamW	4.049 $\pm$ 0.027	3.803 $\pm$ 0.018	4.061 $\pm$ 0.010	3.904 $\pm$ 0.009	3.981 $\pm$ 0.013
	SiLU + Lion	3.993 $\pm$ 0.023	3.744 $\pm$ 0.021	4.001 $\pm$ 0.008	3.848 $\pm$ 0.009	3.921 $\pm$ 0.011
	SiLU + Muon	3.997 $\pm$ 0.030	3.745 $\pm$ 0.017	4.007 $\pm$ 0.013	3.858 $\pm$ 0.010	3.925 $\pm$ 0.013
	SiLU + SOAP	4.096 $\pm$ 0.030	3.863 $\pm$ 0.020	4.114 $\pm$ 0.010	3.957 $\pm$ 0.009	4.035 $\pm$ 0.011
	SiLU + ScheduleFree	4.366 $\pm$ 0.030	4.156 $\pm$ 0.024	4.398 $\pm$ 0.011	4.215 $\pm$ 0.005	4.316 $\pm$ 0.011
	SiLU + CAME	4.368 $\pm$ 0.023	4.150 $\pm$ 0.021	4.406 $\pm$ 0.019	4.249 $\pm$ 0.027	4.330 $\pm$ 0.015
	RLB + AdamW	4.050 $\pm$ 0.030	3.802 $\pm$ 0.021	4.060 $\pm$ 0.011	3.905 $\pm$ 0.008	3.979 $\pm$ 0.010
	RLB + Lion	3.989 $\pm$ 0.029	3.742 $\pm$ 0.021	3.996 $\pm$ 0.011	3.841 $\pm$ 0.008	3.913 $\pm$ 0.014
	RLB + Muon	3.991 $\pm$ 0.026	3.737 $\pm$ 0.019	3.999 $\pm$ 0.011	3.848 $\pm$ 0.005	3.919 $\pm$ 0.015
	RLB + SOAP	4.061 $\pm$ 0.033	3.824 $\pm$ 0.013	4.108 $\pm$ 0.032	3.927 $\pm$ 0.013	4.002 $\pm$ 0.019
	RLB + ScheduleFree	4.361 $\pm$ 0.034	4.142 $\pm$ 0.022	4.386 $\pm$ 0.008	4.212 $\pm$ 0.011	4.308 $\pm$ 0.007
	RLB + CAME	4.450 $\pm$ 0.043	4.225 $\pm$ 0.036	4.480 $\pm$ 0.006	4.267 $\pm$ 0.041	4.365 $\pm$ 0.058

A.1.5 LEARNING-RATE AND WEIGHT-DECAY SENSITIVITY

TODO: Change needed later. Replace the E8 figure, table, and all numerical claims after the corrected E8 runs complete; the current sensitivity results are provisional.

We ran a 100M-token, single-seed sensitivity grid for MatrixPolicy and the two standard SiLU controls. The grid crosses learning rates $\{1, 2, 3, 5\} \times 10^{-4}$ with weight decays 0, 0.05, 0.10, and 0.20 on all five corpora, using the model, tokenizer, evaluation cadence, and token budget in Appendix A.1.2. Let d index a corpus, η denote the outer learning rate, λ the weight-decay coefficient, $h = (\eta, \lambda)$ their matched grid cell, and $L_{d,h,a}(s_k)$ be method a 's validation loss at $s_k \in \mathcal{S} = \{50, 100, \dots, 3050\}$. Writing MP for MatrixPolicy and c for either SiLU control, define the observed-checkpoint margin

$$m_{d,h,c} = \min_{s_k \in \mathcal{S}} [L_{d,h,c}(s_k) - L_{d,h,MP}(s_k)]. \quad (20)$$

A positive margin certifies that MatrixPolicy has lower validation loss at every observed checkpoint from step 50 onward in that matched cell. For a target ℓ , define its sampled first-hit token count as

$$N_{d,h,a}(\ell) = 32,768 \min\{s_k \in \mathcal{S} : L_{d,h,a}(s_k) \leq \ell\}, \quad (21)$$

with $N_{d,h,a}(\ell) = +\infty$ when the set is empty. Therefore $m_{d,h,c} > 0$ implies $N_{d,h,MP}(\ell) \leq N_{d,h,c}(\ell)$ whenever the control's first hit is finite.

Figure 7 gives strict observed-checkpoint dominance in all 80 dataset-grid cells against each control. Every one of the 4,880 paired checkpoint comparisons per control is positive; the smallest observed margin is 0.041 against SiLU with AdamW and 0.045 against SiLU with Muon.

Table 3 makes this target-arrival consequence explicit at the main 100M-token targets from Table 1 and at targets 0.10 and 0.20 loss units lower. Each row partitions all 16 cells and reports mutual non-arrivals as a separate outcome.

At the main targets, MatrixPolicy arrives earlier in all 40 mutual-arrival comparisons per control and is the only method to arrive in 20 additional cells; no cell favors a control. The same ordering holds at both harder target levels. Pooled over mutual arrivals, median token savings at the main, -0.10 , and -0.20 target levels are 15.7%, 19.3%, and 14.3% against SiLU with AdamW and 19.9%, 13.5%, and 10.3% against SiLU with Muon. Node assignments were not matched across the sensitivity sweep, so this analysis reports token arrivals only. The primary-study wall-clock comparison remains in Table 1. Together, the checkpoint and target analyses show the same convergence advantage at all 16 evaluated learning-rate and weight-decay settings for seed 1337.

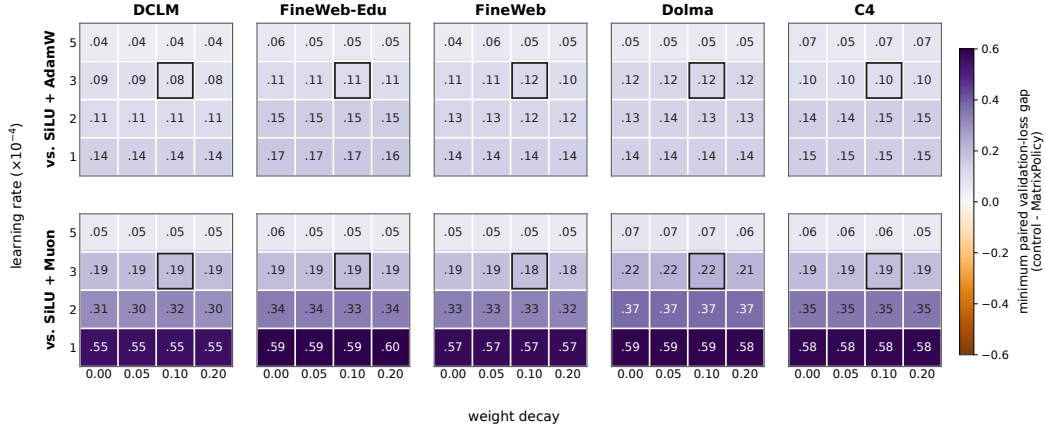


Figure 7: Paired observed-checkpoint margins over the complete learning-rate/weight-decay grid. Each tile is $m_{d,h,c}$ from equation 20 for one dataset, grid cell, and comparator at seed 1337. Positive values mean that MatrixPolicy has lower validation loss at every observed checkpoint from steps 50 through 3050. The dark outline marks the fixed main-study setting (3×10^{-4} , 0.10); all 160 tiles are positive.

Table 3: Complete-grid target arrival for seed 1337 at 100M tokens. Within each dataset, rows give the main target followed by targets 0.10 and 0.20 lower. Each comparator block partitions all 16 cells. “Both reach” lists MatrixPolicy earlier / control earlier / tied, and “only reaches” lists MatrixPolicy / control. Token savings are medians [minimum, maximum] of $100(N_c - N_{MP})/N_c$ over mutual arrivals, where N is first-hit tokens.

Dataset	Target loss	Compared with SiLU + AdamW				Compared with SiLU + Muon			
		Both reach MP / control / tie	Only reaches MP / control	Neither reaches	Token saved (%) median [range]	Both reach MP / control / tie	Only reaches MP / control	Neither reaches	Token saved (%) median [range]
DCLM	4.55	8/0/0	4/0	4	14.4 [7.1,18.4]	8/0/0	4/0	4	19.1 [10.3,29.5]
	4.45	8/0/0	4/0	4	16.9 [8.8,26.5]	7/0/0	5/0	4	11.4 [8.8,40.0]
	4.35	4/0/0	4/0	8	11.9 [11.6,14.0]	4/0/0	4/0	8	10.8 [9.5,11.9]
FineWeb-Edu	4.40	8/0/0	4/0	4	14.4 [10.3,18.4]	8/0/0	4/0	4	19.1 [10.3,27.9]
	4.30	8/0/0	4/0	4	19.6 [11.4,26.5]	8/0/0	4/0	4	23.2 [8.8,37.9]
	4.20	4/0/0	4/0	8	14.0 [14.0,16.3]	4/0/0	4/0	8	9.8 [7.5,12.2]
FineWeb	4.60	8/0/0	4/0	4	16.4 [10.3,22.5]	8/0/0	4/0	4	19.1 [10.3,29.5]
	4.50	8/0/0	4/0	4	21.4 [11.4,28.8]	8/0/0	4/0	4	22.0 [8.8,38.3]
	4.40	4/0/0	4/0	8	15.9 [13.6,15.9]	4/0/0	4/0	8	10.8 [9.5,11.9]
Dolma	4.60	8/0/0	4/0	4	18.1 [10.0,23.3]	8/0/0	4/0	4	22.1 [12.9,34.0]
	4.50	8/0/0	4/0	4	22.5 [13.5,32.8]	4/0/0	8/0	4	13.5 [11.1,13.5]
	4.40	4/0/0	4/0	8	14.3 [13.3,15.2]	4/0/0	4/0	8	13.3 [11.4,13.3]
C4	4.60	8/0/0	4/0	4	14.8 [10.7,18.9]	8/0/0	4/0	4	18.8 [10.7,28.6]
	4.50	8/0/0	4/0	4	19.3 [14.7,25.5]	8/0/0	4/0	4	21.7 [9.4,34.0]
	4.40	4/0/0	5/0	7	14.6 [14.6,16.7]	4/0/0	5/0	7	10.3 [10.3,10.3]

A.2 TODO: LARGER-SCALE TRAINING RESULTS

TODO: Insert larger-scale training details here.

A.3 TODO: COMPONENT ABLATIONS

TODO: Insert component ablation details here.

B RATIONAL LATENT BASIS CALCULUS

The RLB summaries and the reciprocal matrix-pair coordinate follow from the scalar response and the RMS-normalized group map. We suppress layer and group indices throughout this section.

B.1 POLE-FREE RATIONAL RESPONSE

Let $\phi(u)$ collect the nonconstant denominator features. The evaluated P5/Q4 response is

$$\begin{aligned} P(u) &:= \sum_{i=0}^5 a_i u^i, & \phi(u) &:= (|u|, u^2, |u|^3, u^4), \\ Q(u) &:= 1 + \sum_{j=1}^4 |b_j| \phi_j(u), & R(u) &:= \frac{P(u)}{Q(u)}. \end{aligned} \quad (22)$$

Because each denominator feature is nonnegative, $Q(u) \geq 1$ for every $u \in \mathbb{R}$; hence the response is pole-free on $\mathbb{R}$. Away from the kink at $u = 0$,

$$\begin{aligned} R'(u) &= \frac{P'(u)Q(u) - P(u)Q'(u)}{Q(u)^2}, \\ P'(u) &= \sum_{i=1}^5 i a_i u^{i-1}, & Q'(u) &= |b_1| \text{sign}(u) + 2|b_2|u + 3|b_3|u|u| + 4|b_4|u^3. \end{aligned} \quad (23)$$

The selected coefficient derivatives are

$$\frac{\partial R(u)}{\partial a_i} = \frac{u^i}{Q(u)}, \quad \frac{\partial R(u)}{\partial b_j} = -\frac{P(u)}{Q(u)^2} \text{sign}(b_j) \phi_j(u). \quad (24)$$

The implementation selects $\text{sign}(0) = 0$ for both $u = 0$ and $b_j = 0$. At these kinks, the formulas specify the backward derivative chosen by autodiff. For vector inputs, R and R' act coordinatewise. The response, derivative, and coefficient-activity summaries therefore originate from $R(u)$, $R'(u)$, and the parameter gradients associated with equation 24, respectively.

B.2 RMS-NORMALIZED GROUP DIFFERENTIAL

Let m be the group width and let $\epsilon_{\text{rms}} \geq 0$. For $\epsilon_{\text{rms}} > 0$ and $z \in \mathbb{R}^m$, or for $\epsilon_{\text{rms}} = 0$ and $z \neq 0$, define

$$r(z) = \sqrt{m^{-1} \|z\|_2^2 + \epsilon_{\text{rms}}}, \quad u(z) = \frac{z}{r(z)}, \quad h(z) = r(z)R(u(z)). \quad (25)$$

On the domain $r(z) > 0$, direct differentiation gives

$$\text{d}r = \frac{z^\top \text{d}z}{mr} = \frac{u^\top \text{d}z}{m}, \quad \text{d}u = \frac{1}{r} \left(I_m - \frac{uu^\top}{m} \right) \text{d}z. \quad (26)$$

The factor $I_m - uu^\top/m$ is the exact normalized-coordinate differential. It is the tangent projector to the RMS sphere when $\epsilon_{\text{rms}} = 0$. With a positive floor, its radial eigenvalue is $\epsilon_{\text{rms}}/r^2$. Away from scalar-response kinks, the group Jacobian is

$$J_h(z) := \frac{\partial h}{\partial z} = R(u) \frac{u^\top}{m} + \text{diag}(R'(u)) \left(I_m - \frac{uu^\top}{m} \right). \quad (27)$$

This decomposition identifies the scalar factors observed by the live summaries. Response RMS summarizes the radial response factor, and derivative RMS summarizes $\text{diag}(R'(u))$. The complete local Jacobian additionally depends on the normalized-coordinate factor and radial term; the matrix-role gradient also depends on the output matrix and upstream adjoint. The two RMS summaries therefore serve as role-specific scalar proxies.

B.3 POSITIVE PAIR COORDINATE AND FLOOR PERTURBATION

Set $\epsilon_{\text{rms}} = 0$ and write r_0, u_0, h_0 for the unfloored map. For $z \neq 0$ and $\lambda > 0$,

$$r_0(\lambda z) = \lambda r_0(z), \quad u_0(\lambda z) = u_0(z), \quad h_0(\lambda z) = \lambda h_0(z). \quad (28)$$

For a group contribution $y_{l,g} = B_{l,g} h(A_{l,g} x_l)$, with R fixed, $\epsilon_{\text{rms}} = 0$, and $A_{l,g} x_l \neq 0$, the transformation $(A_{l,g}, B_{l,g}) \mapsto (\lambda A_{l,g}, \lambda^{-1} B_{l,g})$ leaves $y_{l,g}$ unchanged. This is the positive pair orbit in equation 3.

The RMS floor perturbs the orbit through both the radius and normalized coordinate. Write

$$r_\epsilon(z) = \sqrt{m^{-1}\|z\|_2^2 + \epsilon_{\text{rms}}}, \quad u_\epsilon(z) = z/r_\epsilon(z), \quad h_\epsilon(z) = r_\epsilon(z)R(u_\epsilon(z)). \quad (29)$$

For $\rho^2 = m^{-1}\|z\|_2^2 > 0$ and $\delta = \epsilon_{\text{rms}}/\rho^2$, define

$$\begin{aligned} s_\epsilon(\lambda, z) &:= \frac{r_\epsilon(\lambda z)}{\lambda r_\epsilon(z)} = \sqrt{\frac{1 + \delta/\lambda^2}{1 + \delta}}, \quad \kappa(\lambda, z) := s_\epsilon(\lambda, z)^{-1}, \\ u_\epsilon(\lambda z) &= \kappa(\lambda, z)u_\epsilon(z). \end{aligned} \quad (30)$$

Fix $c > 0$, and let $\mathcal{I}_c$ contain every scalar coordinate on all segments joining $u_\epsilon(z)$ and $\kappa(\lambda, z)u_\epsilon(z)$ for $\lambda \in [e^{-c}, e^c]$ and all sufficiently small δ . Suppose that R is L_R -Lipschitz on $\mathcal{I}_c$. Reciprocal output compensation then gives, for every λ in this interval, the latent feature bound

$$\begin{aligned} \|\lambda^{-1}h_\epsilon(\lambda z) - h_\epsilon(z)\|_2 &\leq \left| \frac{r_\epsilon(\lambda z)}{\lambda} - r_\epsilon(z) \right| \|R(u_\epsilon(z))\|_2 \\ &\quad + \frac{r_\epsilon(\lambda z)}{\lambda} L_R |\kappa(\lambda, z) - 1| \|u_\epsilon(z)\|_2. \end{aligned} \quad (31)$$

After the pair map, the one-group output difference is $B_{l,g}[\lambda^{-1}h_\epsilon(\lambda z) - h_\epsilon(z)]$, where $B_{l,g}$ denotes the pre-rescale output block. Its norm is at most $\|B_{l,g}\|_{\text{op}}$ times the right-hand side of equation 31. For fixed c and fixed m , uniformly over $\lambda \in [e^{-c}, e^c]$ as $\delta \rightarrow 0$,

$$\begin{aligned} \sup_{\lambda \in [e^{-c}, e^c]} (|s_\epsilon(\lambda, z) - 1| + |\kappa(\lambda, z) - 1|) &= O_c(\delta), \\ \sup_{\lambda \in [e^{-c}, e^c]} \|\lambda^{-1}h_\epsilon(\lambda z) - h_\epsilon(z)\|_2 &= O_c\left(\sqrt{m} [\sup_{\mathcal{I}_c} |R| + L_R] \frac{\epsilon_{\text{rms}}}{\rho}\right). \end{aligned} \quad (32)$$

The first line follows directly from $|s_\epsilon - 1| \leq \frac{1}{2}\delta|\lambda^{-2} - 1|$ and $|\kappa - 1| = |s_\epsilon - 1|/s_\epsilon$. For radii separated from zero and bounded reciprocal moves, the error decays as $\epsilon_{\text{rms}}/\rho$; this rate motivates the clipped log-scale update in MatrixPolicy. The estimate is an away-from-floor statement and gives no vanishing relative-error guarantee as $\rho \rightarrow 0$.

C OPTIMIZER DESIGN AND MATRIXPOLICY GEOMETRY

Let $\text{sg}(\cdot)$ denote stop-gradient. MatrixPolicy first forms detached role summaries, then rescales the current matrix gradients, applies the AdamW/Muon matrix branch, and finally moves the updated pair along the positive coordinate from Appendix B.3. Its effective order is

$$\begin{aligned} \nabla \mathcal{L} &\longrightarrow \text{sg}\left(q^{\text{in}}, q^{\text{out}}, q^R, \pi, \alpha, \hat{d}^{\text{live}}, \hat{o}^{\text{live}}\right) \longrightarrow (\tilde{\nabla} A, \tilde{\nabla} B) \\ &\longrightarrow (A^{\text{desc}}, B^{\text{desc}}) \longrightarrow \text{sg}\left(\hat{d}^{\text{probe}}, \hat{o}^{\text{probe}}, \gamma\right) \\ &\longrightarrow (e^\ell A^{\text{desc}}, e^{-\ell} B^{\text{desc}}). \end{aligned} \quad (33)$$

C.1 LOCAL MATRIX-ROLE GEOMETRY

For one token of one training example, let $\bar{y}_l = \nabla_{y_l} \mathcal{L}$ and $J_{l,g} = \partial h_{l,g} / \partial z_{l,g}$ from equation 27. Here x_l is the RLB input, $z_{l,g} = A_{l,g}x_l$, $h_{l,g} = h(z_{l,g})$, and $y_l = \sum_g B_{l,g}h_{l,g}$. The adjacent matrix gradients factor as

$$\nabla_{B_{l,g}} \mathcal{L} = \bar{y}_l h_{l,g}^\top, \quad \nabla_{A_{l,g}} \mathcal{L} = (J_{l,g}^\top B_{l,g}^\top \bar{y}_l) x_l^\top. \quad (34)$$

Batch and sequence gradients are sums or averages of these outer-product terms. The output-role gradient contains the realized feature $h_{l,g}$, whereas the input-role gradient contains the local Jacobian $J_{l,g}$. This factorization supports response RMS and derivative RMS as scalar role proxies. The full gradient retains the omitted radius, normalized-coordinate factor, matrix orientation, data distribution, and upstream adjoint.

The pressure summaries quantify relative gradient scale. For a pure gradient step $W^+ = W - \eta \mathcal{G}$, with $W \neq 0$ and $\eta \|\mathcal{G}\|_F \leq c \|W\|_F$ for fixed $c < 1$, Taylor expansion gives

$$\log \text{rms}(W^+) - \log \text{rms}(W) = -\eta \frac{\langle \mathcal{G}, W \rangle_F}{\|W\|_F^2} + O_c \left(\eta^2 \frac{\|\mathcal{G}\|_F^2}{\|W\|_F^2} \right), \quad (35)$$

while Cauchy–Schwarz yields

$$\left| \frac{\langle \mathcal{G}, W \rangle_F}{\|W\|_F^2} \right| \leq \frac{\|\mathcal{G}\|_F}{\|W\|_F} = \frac{\text{rms}(\mathcal{G})}{\text{rms}(W)}. \quad (36)$$

Thus $\text{rms}(\mathcal{G})/\text{rms}(W)$ bounds the magnitude of the first-order log-RMS change for a pure gradient step. The same statement has an exact floored counterpart. If n is the number of entries of W , then

$$\left| \frac{d}{d\eta} \log \text{rms}_{\epsilon_p}(W - \eta \mathcal{G}) \right|_{\eta=0} = \frac{|\langle \mathcal{G}, W \rangle_F|}{\|W\|_F^2 + n\epsilon_p} \leq \frac{\text{rms}_{\epsilon_p}(\mathcal{G})}{\text{rms}_{\epsilon_p}(W)}. \quad (37)$$

The implemented p^{in} and p^{out} are the right-hand side of equation 37, computed from post-global-clipping gradients before the group policy. They summarize unsigned pre-policy gradient scale; AdamW and Muon subsequently construct signed, state-dependent updates. Their EMA contrast $\pi_{l,g}$ identifies the role with larger recent relative gradient scale. The activity contrast $\alpha_{l,g}$ compares the block-weighted rational-coefficient gradient memory with the geometric mean of the two matrix-pressure memories and supplies a calibrated policy signal for relative coefficient activity.

For a differentiable data loss whose dependence on $(A_{l,g}, B_{l,g})$ is only through the RLB block output, and at representatives with $A_{l,g}x_l \neq 0$ for the evaluated samples, the unfloored fixed-response orbit satisfies

$$\left. \frac{d}{ds} \mathcal{L}(e^s A_{l,g}, e^{-s} B_{l,g}) \right|_{s=0} = \langle \nabla A_{l,g}, A_{l,g} \rangle_F - \langle \nabla B_{l,g}, B_{l,g} \rangle_F = 0. \quad (38)$$

The final control follows this zero-loss tangent in the idealized map to alter relative matrix pressure.

C.2 CENTERED GROUP CONTROL

The first control is most transparent in log space. Define the layerwise mean log gains

$$\bar{d}_l = \frac{1}{G} \sum_{j=1}^G \log \hat{d}_{l,j}^{\text{live}}, \quad \bar{o}_l = \frac{1}{G} \sum_{j=1}^G \log \hat{o}_{l,j}^{\text{live}}. \quad (39)$$

The gain factors in equation 12 obey

$$\log c_{l,g,\text{in}}^{\text{gain}} = -0.20 \phi_t (\log \hat{d}_{l,g}^{\text{live}} - \bar{d}_l), \quad \log c_{l,g,\text{out}}^{\text{gain}} = -0.20 \phi_t (\log \hat{o}_{l,g}^{\text{live}} - \bar{o}_l). \quad (40)$$

Above-geometric-mean derivative or response gain therefore reduces the corresponding role multiplier. The pressure factors act with opposite signs,

$$\log c_{l,g,\text{in}}^{\text{pressure}} = -0.10 \phi_t \tilde{\pi}_{l,g}, \quad \log c_{l,g,\text{out}}^{\text{pressure}} = +0.10 \phi_t \tilde{\pi}_{l,g}, \quad (41)$$

so positive $\pi_{l,g}$ reduces the input-role gradient and increases the output-role gradient before recentering. The shared activity factor is

$$\log c_{l,g}^{\text{activity}} = -0.20 \phi_t \text{ReLU} \left(\frac{\alpha_{l,g} - 0.05}{0.45} \right). \quad (42)$$

To distinguish the stages of the implementation, let

$$c_{l,g,\sigma}^{\text{pre}} := c_{l,g,\sigma}^{\text{gain}} c_{l,g,\sigma}^{\text{pressure}} c_{l,g}^{\text{activity}}, \quad \hat{c}_{l,g,\sigma} := \frac{c_{l,g,\sigma}^{\text{pre}}}{\text{geomean}_{j=1}^G c_{l,j,\sigma}^{\text{pre}}}, \quad c_{l,g,\sigma} := \text{clip}(\hat{c}_{l,g,\sigma}, 0.75, 1.35), \quad (43)$$

$$\frac{1}{G} \sum_{g=1}^G \log \hat{c}_{l,g,\sigma} = 0.$$

The zero-mean identity holds exactly before the final clip; the clip may shift the layerwise mean log multiplier. Any factor common to every group in one role cancels during recentering, so the final gate reflects groupwise differences in the combined log control. It preserves each group-gradient direction and multiplies its magnitude by a positive scalar before AdamW and Muon construct their state-dependent update directions.

C.3 TRANSIENT MUON MATRIX STEP

The factors $s_{l,\sigma}^{\text{Adam}}$ in equation 14 scale the AdamW branch, while $\varrho_\sigma(l)$ also enters the Muon fraction in equation 15. Their combination can therefore change both the magnitude and direction of the sequential matrix update. AdamW maintains entrywise moments (Kingma & Ba, 2015; Loshchilov & Hutter, 2019),

$$\begin{aligned} M &\leftarrow \beta_1 M + (1 - \beta_1)G, \quad V \leftarrow \beta_2 V + (1 - \beta_2)(G \odot G), \\ D_{ij}^{\text{Adam}} &= \frac{M_{ij}}{\sqrt{V_{ij} + \epsilon}}, \end{aligned} \quad (44)$$

where the display suppresses bias correction and decoupled weight decay. This normalization acts coordinatewise; the subsequent Muon branch controls the singular spectrum of the matrix direction.

Matrix spectra matter for the RLB pair because equation 34 gives single-example adjacent gradients as outer products; after averaging over batch and sequence positions, the training gradients are sums of such terms. Singular modes of B_l couple output-residual directions to latent features, and singular modes of A_l couple latent coordinates to input-residual directions. During early basis formation, a highly anisotropic singular spectrum concentrates movement in a narrow set of these coupled directions.

Let $H_{l,\sigma,t} = U\Sigma V^\top$ be the compact SVD of the tensor produced after group-gradient scaling and Muon’s momentum/Nesterov construction. Its ideal polar direction is

$$\mathcal{P}(H_{l,\sigma,t}) = UV^\top, H = \sum_i \sigma_i u_i v_i^\top \mapsto \sum_i u_i v_i^\top \quad \text{on the nonzero singular subspace.} \quad (45)$$

The polar map preserves the row and column singular vectors, replaces every nonzero singular value by one, and retains the nonzero singular subspace of $H_{l,\sigma,t}$. The polar factor solves

$$\mathcal{P}(H) \in \arg \max_{\|D\|_2 \leq 1} \langle D, H \rangle_F, \quad \langle \mathcal{P}(H), H \rangle_F = \|H\|_*, \quad (46)$$

$$\mathcal{P}(aH) = \mathcal{P}(H) \quad (a > 0, H \neq 0).$$

The implementation instead applies a finite polynomial map. Let $\bar{H}_{l,\sigma,t}$ be the bfloat16-rounded $H_{l,\sigma,t}$, transposed when needed so that its row dimension is no larger than its column dimension. It sets

$$\begin{aligned} X_0 &= \frac{\bar{H}}{\max\{\|\bar{H}\|_F, 10^{-7}\}}, \\ X_{k+1} &= 3.4445X_k - 4.775(X_k X_k^\top)X_k + 2.0315(X_k X_k^\top)^2 X_k, \quad k = 0, \dots, 4, \\ D_{l,\sigma,t}^\mu &= X_5, \end{aligned} \quad (47)$$

and transposes the result back. Ignoring subsequent finite-precision roundoff, if $\bar{H} = \bar{U}\bar{\Sigma}\bar{V}^\top$ and $s = \max\{\|\bar{H}\|_F, 10^{-7}\}$, then

$$X_5 = \bar{U} f^{\circ 5}(\bar{\Sigma}/s) \bar{V}^\top, \quad f(r) = 3.4445r - 4.775r^3 + 2.0315r^5. \quad (48)$$

Thus the real-arithmetic recurrence preserves the singular vectors of the rounded input and changes its singular values through $f^{\circ 5}$. The nonmonotone finite polynomial has no universal condition-number or singular-spread reduction guarantee. The implementation additionally uses match-RMS learning-rate adjustment (Liu et al., 2025).

The branch equation in equation 16 is sequential because AdamW and Muon keep separate states. Let $\mathcal{D}_{l,\sigma,t}^{\text{Adam}}$ and $\mathcal{D}_{l,\sigma,t}^\mu$ denote their complete first-order directions, including bias correction, decoupled weight decay, and, for Muon, the finite Newton–Schulz and match-RMS adjustments. The sequential step has expansion

$$\Delta M_{l,\sigma} = -\eta_t s_{l,\sigma}^{\text{Adam}} (1 - \mu_{l,\sigma,t}) \mathcal{D}_{l,\sigma,t}^{\text{Adam}} - \eta_t s_\mu \mu_{l,\sigma,t} \mathcal{D}_{l,\sigma,t}^\mu + O(\eta_t^2), \quad (49)$$

as $\eta_t \rightarrow 0$ with the current gradients, optimizer states, schedules, and matrix shapes fixed; the hidden constant depends on those quantities. This is the first-order expansion after the group control has set the role-specific gradient scales. The polynomial reweights the existing singular modes without creating new singular directions. MatrixPolicy confines this intervention to the early window in equation 15; AdamW and reciprocal pair balancing remain after the Muon fraction reaches zero.

The Muon gate in equation 15 is monotone in the two layer summaries. Away from clipping boundaries and the ReLU kink, the derivatives of $\log \psi_{l,t}$ with respect to $\Psi_{l,t}$ and $\Omega_{l,t}$ are -0.30 and -0.65 inside the ψ clipping interval. Thus the Muon fraction is largest when role pressure and rational-coefficient activity are both small. Large $|\pi_{l,g}|$ marks unequal input/output pressure, and large $\alpha_{l,g}$ marks rational coefficients moving strongly relative to the matrix pair. These are precisely the regimes where the Muon substep is attenuated.

C.4 RECIPROCAL PAIR BALANCING

After the descent branch, MatrixPolicy applies the reciprocal reparameterization

$$A_{l,g}^+ = e^{\ell_{l,g,t}} A_{l,g}^{\text{desc}}, \quad B_{l,g}^+ = e^{-\ell_{l,g,t}} B_{l,g}^{\text{desc}}, \quad (50)$$

every five optimizer steps. The construction follows the positive coordinate in Appendix B.3. With the rational response held fixed, the unfloored map satisfies $h_{l,g}(\lambda A_{l,g} x) = \lambda h_{l,g}(A_{l,g} x)$ for $\lambda > 0$, so replacing $(A_{l,g}, B_{l,g})$ by $(\lambda A_{l,g}, \lambda^{-1} B_{l,g})$ leaves $B_{l,g} h_{l,g}$ unchanged. The floor-error bound in equation 31 quantifies the departure from this identity.

For the coordinate analysis below, write $(A_{l,g}, B_{l,g})$ for a generic post-descent pair and $(A_{l,g}^+, B_{l,g}^+)$ for its reciprocal rescaling. The loss is assumed differentiable and to depend on this pair only through the RLB block output, with $A_{l,g} x_l \neq 0$ in the unfloored map.

Equivalent unfloored representatives have different relative gradient pressures. Let

$$p_{l,g}^A = \frac{\text{rms}(\nabla A_{l,g})}{\text{rms}(A_{l,g})}, \quad p_{l,g}^B = \frac{\text{rms}(\nabla B_{l,g})}{\text{rms}(B_{l,g})}. \quad (51)$$

In the fixed-response, unfloored map, $J_h(e^\ell z) = J_h(z)$ almost everywhere. Evaluating the instantaneous gradients at the transformed representative gives

$$\begin{aligned} \nabla_{A_{l,g}^+} \mathcal{L} &= e^{-\ell_{l,g,t}} \nabla_{A_{l,g}} \mathcal{L}, & \nabla_{B_{l,g}^+} \mathcal{L} &= e^{\ell_{l,g,t}} \nabla_{B_{l,g}} \mathcal{L}, \\ p_{l,g}^{A,+} &= e^{-2\ell_{l,g,t}} p_{l,g}^A, & p_{l,g}^{B,+} &= e^{2\ell_{l,g,t}} p_{l,g}^B. \end{aligned} \quad (52)$$

The law in equation 52 concerns freshly evaluated, unfloored pressures before global clipping. Applying only the numerical floor in the pressure measurement to this ideal covariance law, and writing $a = \text{rms}(A)^2$, $b = \text{rms}(B)^2$, $g_A = \text{rms}(\nabla A)^2$, and $g_B = \text{rms}(\nabla B)^2$, gives

$$p_{\epsilon_p}^{A,+} = \frac{\sqrt{e^{-2\ell} g_A + \epsilon_p}}{\sqrt{e^{2\ell} a + \epsilon_p}}, \quad p_{\epsilon_p}^{B,+} = \frac{\sqrt{e^{2\ell} g_B + \epsilon_p}}{\sqrt{e^{-2\ell} b + \epsilon_p}}. \quad (53)$$

A global clipping factor may further scale both numerators, and the EMA memories retain their temporal values. The activation RMS floor also perturbs the fresh gradient covariance, as quantified at the function level in Appendix B.3; equation 53 isolates the separate effect of the pressure-measurement floor. In the ideal covariance law, positive ℓ gives the corrective raw-pressure direction by reducing the input-side pressure and increasing the output-side pressure. The $+0.25\pi_{l,g}$ term in equation 17 therefore biases $\tau_{l,g}$ in that direction when the input role has larger recent pressure.

The curve-shape target follows from a normalized separable surrogate. Holding positive data- and orientation-dependent factors $C_{l,g}^A, C_{l,g}^B > 0$ fixed within one group gives the general form

$$\tilde{p}_{l,g}^A(\gamma) = C_{l,g}^A \hat{d}_{l,g}^{\text{probe}} e^{-\gamma}, \quad \tilde{p}_{l,g}^B(\gamma) = C_{l,g}^B \hat{o}_{l,g}^{\text{probe}} e^{\gamma}. \quad (54)$$

Their unique equality point is

$$\gamma_{l,g}^{\text{bal}} = \frac{1}{2} \left(\log \hat{d}_{l,g}^{\text{probe}} - \log \hat{o}_{l,g}^{\text{probe}} + \log C_{l,g}^A - \log C_{l,g}^B \right), \quad (55)$$

MatrixPolicy uses the symmetric normalization $C_{l,g}^A = C_{l,g}^B$, which sets the unknown relative nuisance term to zero and yields the first term of $\tau_{l,g}$. This is a balance coordinate for the scalar surrogate; full matrix gradients retain the data and orientation factors summarized by $C_{l,g}^A, C_{l,g}^B$.

For nonzero matrices and unfloored RMS values,

$$\begin{aligned}\gamma_{l,g} &= \log \text{rms}(A_{l,g}) - \log \text{rms}(B_{l,g}), \\ \gamma_{l,g}^+ &= \gamma_{l,g} + 2\ell_{l,g,t}.\end{aligned}\tag{56}$$

Substituting the unclipped log step from equation 18 gives

$$\begin{aligned}\zeta_{l,g,t} &= s_{l,t} \chi_{l,g}^{\text{rs}}, \\ \gamma_{l,g}^+ &= (1 - \zeta_{l,g,t})\gamma_{l,g} + \zeta_{l,g,t}\tau_{l,g}.\end{aligned}\tag{57}$$

Since $\delta_l \in [0, 1]$, the evaluated depth factor lies in $[0.925, 1.075]$. Hence $0 \leq s_{l,t} \leq 0.5375$ and $\chi_{l,g}^{\text{rs}} \leq e^{0.125}$, so $0 \leq \zeta_{l,g,t} < 0.61$. The idealized unclipped coordinate is therefore a strict contraction toward $\tau_{l,g}$ whenever $\zeta_{l,g,t} > 0$. With the 0.030 log-step clip, the unfloored error obeys

$$|\gamma_{l,g}^+ - \tau_{l,g}| = |\gamma_{l,g} - \tau_{l,g}| - \min\{\zeta_{l,g,t}|\gamma_{l,g} - \tau_{l,g}|, 0.060\}.\tag{58}$$

Thus clipping preserves one-step progress but replaces a uniform linear contraction by a bounded additive move.

The implemented γ uses rms_{ϵ_p} . If $a = \text{rms}(A_{l,g})^2$ and $b = \text{rms}(B_{l,g})^2$ immediately before rescaling, its exact change is

$$\begin{aligned}\gamma_{\epsilon_p}^+ - \gamma_{\epsilon_p} &= \frac{1}{2} \log \frac{e^{2\ell}a + \epsilon_p}{a + \epsilon_p} - \frac{1}{2} \log \frac{e^{-2\ell}b + \epsilon_p}{b + \epsilon_p}, \\ \frac{d\gamma_{\epsilon_p}^+}{d\ell} &= \frac{e^{2\ell}a}{e^{2\ell}a + \epsilon_p} + \frac{e^{-2\ell}b}{e^{-2\ell}b + \epsilon_p} \in [0, 2).\end{aligned}\tag{59}$$

The floor therefore preserves the movement direction but attenuates it near zero; when both blocks vanish, the floored log ratio does not move.

The activity factor has the opposite sign in this speed control. Large $\alpha_{l,g}$ damps gradient-driven matrix motion in equation 12 and equation 15; here $\chi_{l,g}^{\text{rs}} = \exp(0.10\bar{\alpha}_{l,g})$ controls the speed of the post-descent pair move, while the group and Muon controls determine gradient-driven matrix motion. In the reported implementation the wrapper rescales $A_{l,g}$ and $B_{l,g}$ after the descent branch, while the AdamW first- and second-moment buffers and Muon momentum buffer remain in their pre-rescale coordinates. The contraction in equation 57 therefore describes the parameter-pair coordinate, rather than the complete optimizer state.